

COURSE SYLLABUS

ECE341 – Introduction to Computer Hardware

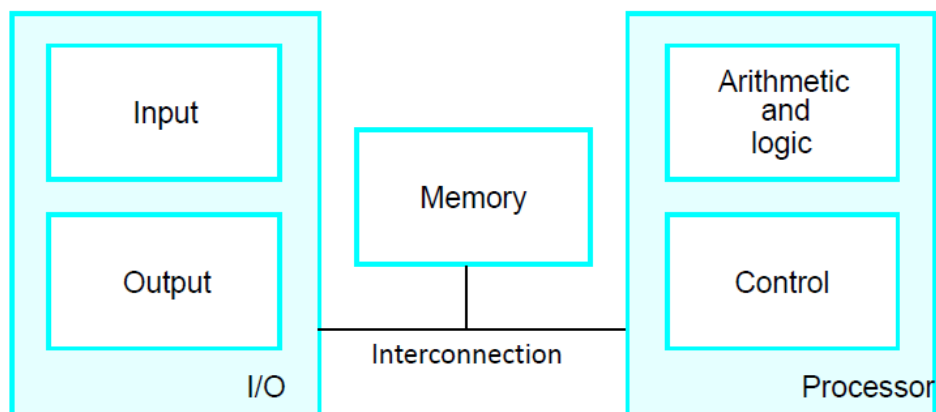
1. GENERAL INFORMATION

Course name:	Introduction to Computer Hardware
Course name (in Vietnamese):	
Course ID:	ECE341
Knowledge block:	
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	CS 201 System Programming
Prior-course:	
Instructors:	A.Prof. Dinh Dien, ddien@fit.hcmus.edu.vn

2. COURSE DESCRIPTION

An overview of computer architecture and programming from a hardware viewpoint. Topics covered include: digital logic; arithmetic operations; pipelining; CISC/RISC; datapath design ;memory hierarchy; virtual memory; input/output techniques; computer system components.

End Goal: To understand the hardware implementation of different components of a modern computer



Basic functional units of a computer

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Apply digital logic gates, flip flops to build a function block.	
G2	Describe basic functional units of computer.	
G3	Conduct a basic computer architecture – data path, control, and buses	
G4	Describe CISC & RISC architectures	

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Use basic logic gate to build encoder, decoder, Mux, deMux	I
G1.2	Use logic gate and flip flop to build counter, Shift register	I
G2.1	Explain and demonstrate ALU, Register files, Memory	T
G3.1	Construct Simple RISC datapath	T
G4.1	Practice Assembly language programming for CISC and RISC CPU	T

5. TEACHING PLAN

THEORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)
1	Digital logic design Sequential logic	Understanding and applying	Appendix A (Sections A.1, A.2, A.3, and A.4, excluding A.3.1 and A.3.2) Appendix A (Section A.6)
2	Registers, Counters, FSM Decoders, Multiplexers, PLD	Understanding and applying	Appendix A (Sections A.7, A.8, A.13) Appendix A (Sections A.9, A.10, A.11, A.12)
3	Computer Arithmetic	Understanding	Chapter 9 (Section 9.1)
4	Fast Adders, Multipliers	Understanding and applying	Chapter 9 (Section 9.2) Chapter 9 (Sections 9.3, 9.4, 9.5.1)
5	Basic Organization of Computers	Understanding	Chapter 1 (Section 1.2 and 1.6)
6	Basic Processing Unit	Understanding and applying	Chapter 5
7	Pipeline	Understanding	Chapter 6
8	Memory System	Understanding and applying	Chapter 8
9	Cache Memory	Understanding	Chapter 8
10	Basic Input/Output Direct Memory Access (DMA)	Understanding	Chapter 3 (Sections 3.1, 3.2.1, 3.2.3)

			Chapter 8 (Section 8.4)
11	Final Exam		

LABORATORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Introduction to Logisim, digital logic gate	G1.1, G1.2	Explain and demonstrate	LW1
2	Encoder, Decoder, Multiplexer	G1.1, G1.2	Explain and demonstrate	LW2
3	Counter	G1.2	Lecturing Provide instructions Demonstration, discussion	LW3
4	FSM	G1.2, G2.1	Lecturing Explain and demonstrate Q&A	LW4
5	ALU design	G2.1	Explain and demonstrate	LW5
6	Register files	G2.1	Case study and discussion	LW6
7	Basic RISC CPU	G4.1	Lecturing Demonstration	LW7

8	CISC CPU I/O programming	G3.1, G4.1	Provide instructions Demonstration, discussion	LW8
9	RISC CPU Assembly	G3.1, G4.1	Provide instructions Demonstration, discussion	LW9
10	RISC CPU Assembly	G3.1, G4.1	Provide instructions Demonstration, discussion	LW9

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Projects and Lab exercise	Weekly labwork		25%
A2	Exams			
A21	Midterm Exam	Written (closed book) Understanding and solving problems based on topics: logic gate, encoder, decoder, finite state machine, counter.	G1.2, G1.2, G2.1	25%
A22	Final Exam	Written (closed book) Understanding and solving problems based on topics: Memory, Registerfiles,	G2.1, G3.1, G4.1	50%

		RISC CPU datapath, RISC assembly.		
A31	Bonus	Is a way to raise your grade. It is granted one point at a time for good ideas expressed in class, for constructive contributions, identification of errors in lectures, handouts, and homework assignments.		unlimited

7. RESOURCES

Textbooks

- Computer Organization and Embedded Systems, 6th Edition, Hamacher, Vranesic, Zaky and Manjikian. McGraw-Hill, 2011. ISBN 9780073380650 / 0073380652

Other

- Logic and computer design fundamentals, 5th Edition, Mano, Kime and Martin. Pearson, 2013. ISBN 9780133760637/0133760634
- Computer Organization and Design: the Hardware software interface, RISC-V Edition, Morgan Kaufmann, 2017. ISBN 9780128122754/0128122757

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.

- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.
- Online:
 - Log-in to the Zoom is required to join the class
 - The account name is set according to the syntax:
[Student ID] [Full name]
For example: 18125510 Nguyễn Thị Minh Ánh
 - Activities: Answering questions for adding points requires #bonus before answers.
 - Students must ensure the same time as scheduled (On time - Punctuality - Attendance).